Dongwook **Kwon**

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Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization, Multi-Agent Systems (MAS), and AI-driven anomaly detection for critical infrastructure. Demonstrated research excellence through co-first authored publications in top-tier peer-reviewed venues, including an Oral presentation at EMNLP 2025 and a paper at ACL 2026 (both Industry Tracks), developed in collaboration with LG Electronics. Inventor of multiple registered patents in network and medical device anomaly detection. A core engineer who engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company. With foundational research experience in nuclear cybersecurity and medical cyber-physical systems, poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Apr 2026	**Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984	*Republic of Korea*
Oct 2025	**System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM**, KR 10-2025-0160217 (Pending)	*Republic of Korea*
Aug 2025	**GPT API-Based Network Anomaly Detection**, KR 10-2848814	*Republic of Korea*
Aug 2024	**Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof**, KR 10-2024-0110559 (Pending)	*Republic of Korea*

Publications

Journal Articles

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

- R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026.
  †Equal contribution.
- J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
  †Equal contribution.
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. ***Best Poster Award — Silver.***
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. ***Outstanding Paper Award.***

Research & Professional Experience

Suresoft Technologies — *Seongnam-si, Korea*

AI Engineer, Intelligent Agent Technology Team — *Dec. 2025 - Present*

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University — *Seoul, Korea*

Research Assistant — *Aug. 2021 - Dec. 2025*

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada (with University of Toronto) — *Toronto, ON, Canada*

Visiting Researcher — *Jan. 2025 - Jul. 2025*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego — *San Diego, CA, USA*

AI Development Project Participant, Summer Program — *Jul. 2022 - Aug. 2022*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University — *Seoul, Korea*

Master of Science in Computer Engineering — *Mar. 2024 - Feb. 2026*

- GPA: 4.33/4.5 (98.1%)

University of Toronto — *Toronto, ON, Canada*

Graduate Research Program in Mechanical and Industrial Engineering — *Jan. 2025 - Jul. 2025*

- GPA: 3.68/4.0

Kwangwoon University — *Seoul, Korea*

Bachelor of Science in Electronics and Communications Engineering — *Mar. 2019 - Feb. 2024*

- GPA: 3.41/4.5 (88.1%)

Honors & Awards

Major Awards

Aug. 2022	**Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego	*United States*
Dec. 2021	**Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries	*South Korea*

Academic Recognition

Nov. 2025	**Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers	*South Korea*
Nov. 2024	**Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)	*South Korea*
Nov. 2023	**Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering	*South Korea*